Project Components Explained

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1. LangGraph:

langGraph is an implementation of DCG (directed cyclic graphs) as a data structure, where the langGraph acts as a state machine on each node of the graph (a state machine is a machine that stays only at a single state at any given time).

so langGraph implements these DCG as decision trees to follow by Ai agents

2. LLM: A Large Language model is a neural network pre-trained on a large corpus of data to output statistically generated similar data.

3. ReAct Agent Loop: ReAct (Reason + Act) is the way that Ai agents work.

more accurately it works as reason -> act -> observe -> repeat, where it first reasons the task i.e. what it needs to do, then it acts i.e. uses ai tools to work on the native data, then it observes the changes to update it's internal state and finally it loops back to reason

4. Planning tool: This acts as the deterministic program that breaks a given complex task into simpler sub-tasks.

5. Virtual File System Tools: These tools are what is used to do the work in the ReAct loop, in order to natively work on the operating system as if a human was doing it instead of an Ai. However as it is communication between applications, Ai operates on a machine level with the OS.

6. Sub-Agent Delegation Tool: This is another deterministic program that decides which sub-tasks are assigned to which Ai sub-agents for specific sub-tasks to be done easily in the specific manner the sub-Ai was pre-trained on.

7. State Graph: State graph is the snapshot of the evolving state of the task in between the nodes of the aforementioned DCG as it's getting worked on by the Ai agent.

8. Standard Tools: Native human tools that can be used by Ai as is, like cmd prompt.

9. Detailed System Prompt: This sets the exact state of the system to known by the Ai before the Act part of the ReAct loop, It also grounds the Ai to the status of the system, so that it can differentiate the before and after the Act in ReAct.

10. LangSmith: it's a platform to primarily evaluate any Ai agent and provides tools to debug, test, engineer etc. on the Ai agent.